

Introduction to Decision-Focused Learning for airlines

Part 1: delay prediction, stochastic planning, and column generation

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ÉCOLE NATIONALE DES
PONTS
ET CHAUSSÉES



IP PARIS

AIRFRANCE 

Agenda for the day

Part 1: OR

1. The practical problem
2. Classical OR: column generation
3. Practice session 1: OR for tail assignment in practice

Part 2: DFL

4. Intro to Decision-Focused Learning
5. Demo on a toy problem
6. Practice session 2: DFL for tail assignment in practice

Objectives:

- Use as little math/theory as possible
- Showcase a generic method that can be applied to a wide range of problems
- Intuition about how and why it works
- Demonstrate how to apply it to an airline problem

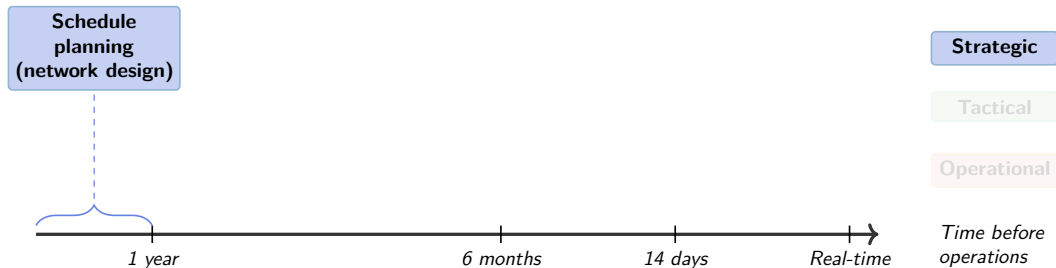
Part 1: outline

- 1 The stochastic tail assignment problem
- 2 Modeling delays
- 3 Optimizing with column generation
- 4 Practice session 1

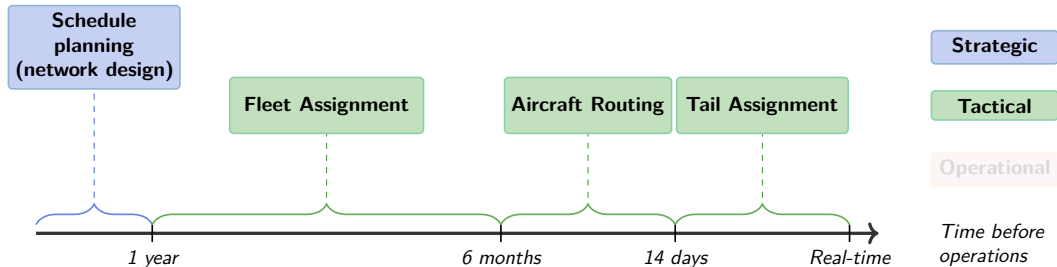
Overview

- 1 The stochastic tail assignment problem
- 2 Modeling delays
- 3 Optimizing with column generation
- 4 Practice session 1

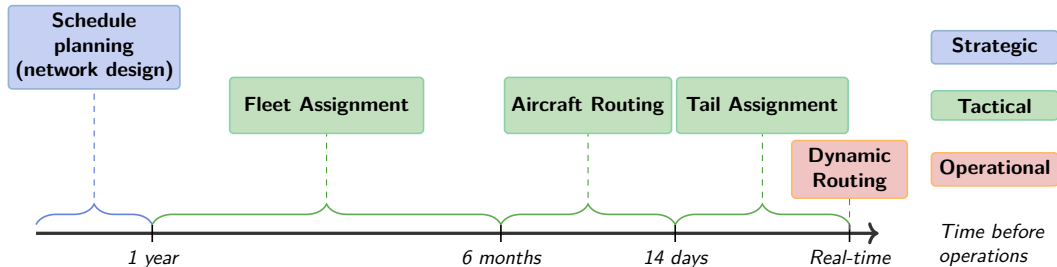
Airline operations at Air France



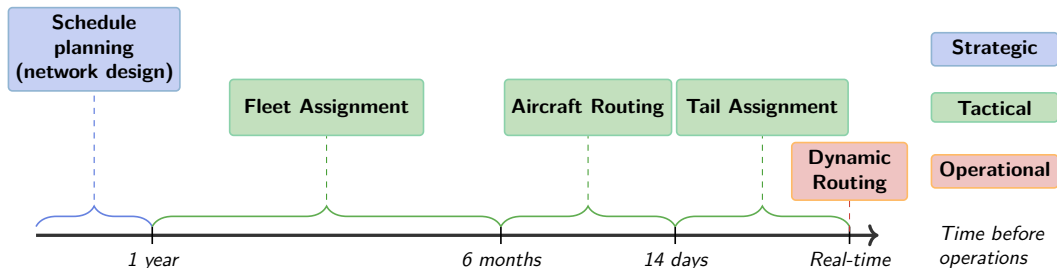
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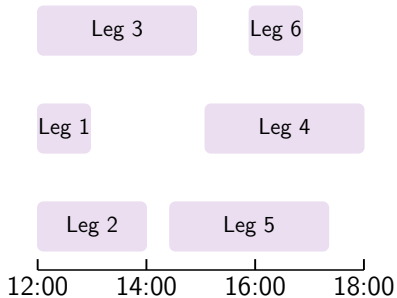


Airline operations at Air France



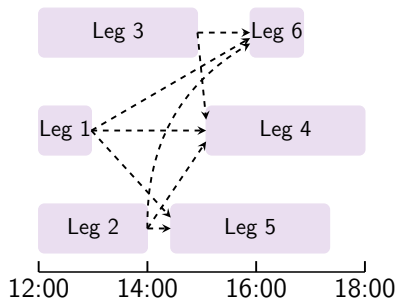
- Focus on **tail assignment**
- Generic methodology, can be applied to other problems

The (deterministic) tail assignment problem



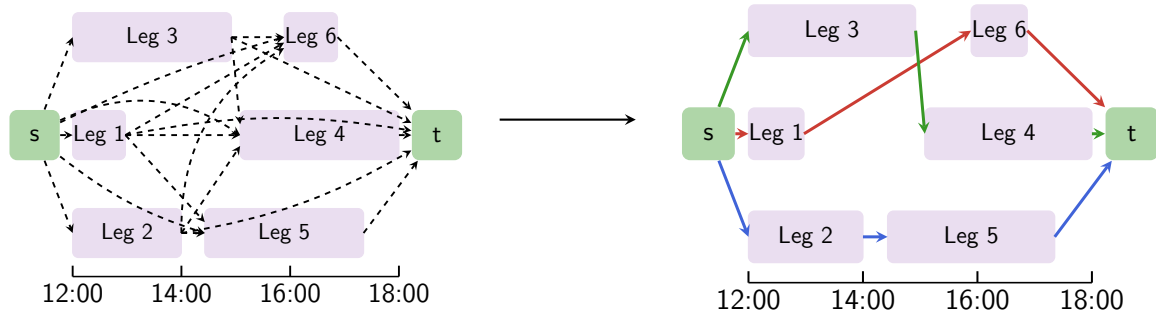
Flight Leg	Departure	Arrival
1: B→A	12:00	13:00
2: C→A	12:00	14:00
3: D→A	12:00	15:00
4: A→D	15:10	18:10
5: A→C	14:30	17:30
6: A→B	16:00	17:00

The (deterministic) tail assignment problem

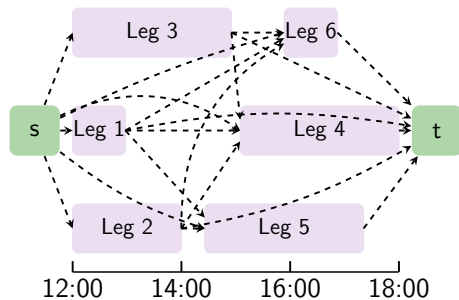


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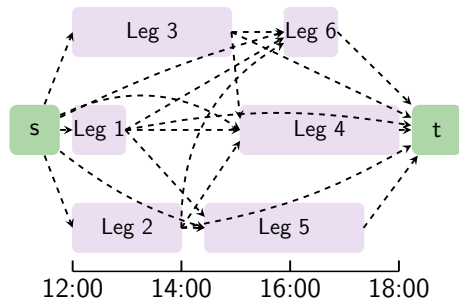


Aircraft	Fuel Factor
1	1.0
2	2.0
3	3.0

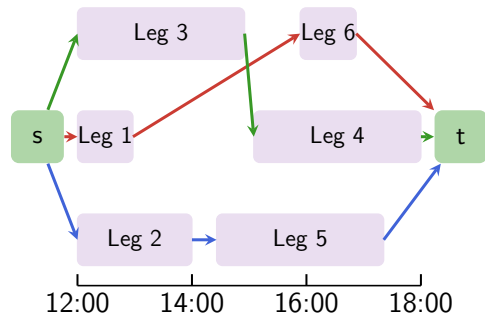
$\min_{\text{routes } y}$ $\overbrace{\text{fuel cost} + \text{connection cost}}^{\text{operational cost}}$

subject to $\left\{ \begin{array}{l} \text{valid routes,} \\ \text{maintenance constraints,} \\ \text{mandatory connections.} \end{array} \right.$

The (deterministic) tail assignment problem



$$\min_{y \in \mathcal{Y}(x)} \theta^\top y$$

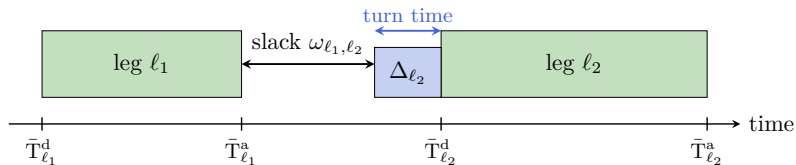


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- **Deterministic** problem: easily solved by commercial MILP solvers

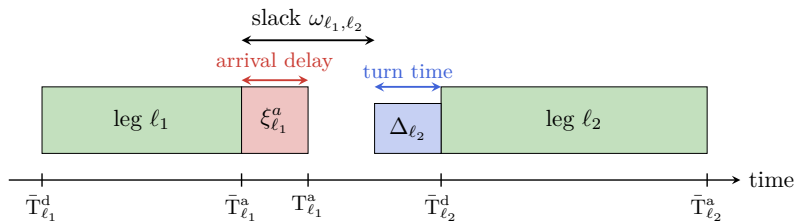
Delay propagation

- A connection along an aircraft route:



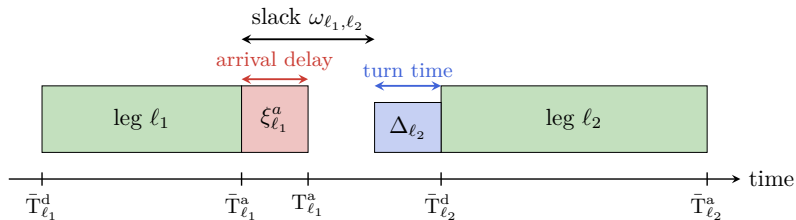
Delay propagation

- No delay propagation:

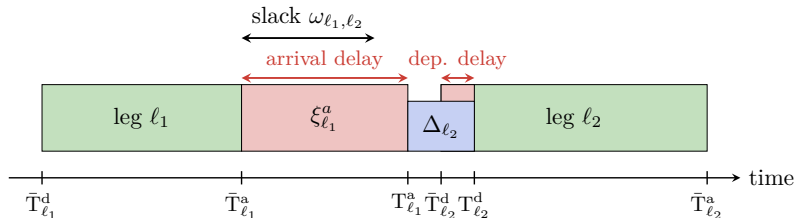


Delay propagation

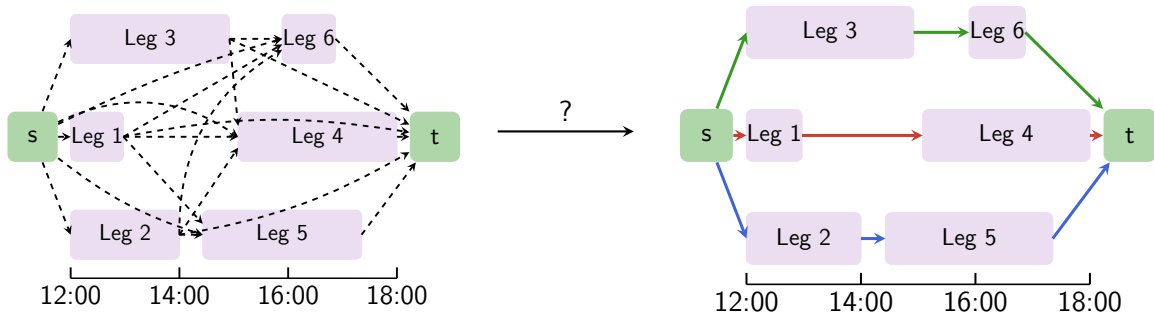
■ No delay propagation:



■ Delay propagation:



The **stochastic** tail assignment problem



- **Deterministic** problem: easily solved by commercial MILP solvers
- Does not take **delays** into account, minimum turn time is not enough
- **Stochastic** problem: non-linear

The **stochastic** tail assignment problem

$\min_{\text{routes } y}$ operational cost + $\mathbb{E}_{\text{delays}}[\text{delays cost}]$

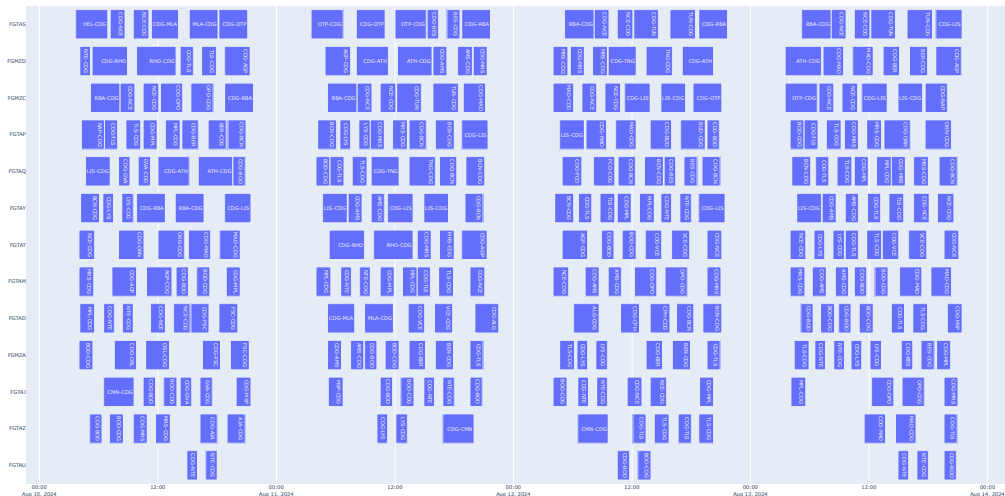
subject to $\left\{ \begin{array}{l} \text{valid routes,} \\ \text{maintenance constraints,} \\ \text{mandatory connections.} \\ \text{etc.} \end{array} \right.$

■ Delays cost:

- ▶ customer dissatisfaction and compensation
- ▶ additional operational costs

Can we leverage historical data and recent advances in machine learning to become data-driven and scale up?

How to scale on large-scale industrial instances for stochastic problems ?



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Delay propagation along aircraft routes

The diagram illustrates the decomposition of arrival delay into its components. It features the equation $\xi_\ell^a = \xi_\ell^d + \varepsilon_\ell^a$ where each term is enclosed in a colored box: ξ_ℓ^a is in a light green box, ξ_ℓ^d is in a light green box, and ε_ℓ^a is in a light red box. A green arrow labeled "Arrival delay" points to the ξ_ℓ^a box. A green arrow labeled "Departure delay" points to the ξ_ℓ^d box. A red arrow labeled "Arrival root delay" points to the ε_ℓ^a box.

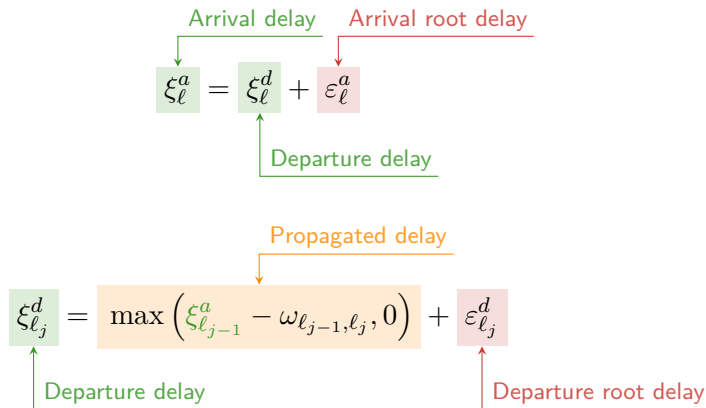
$$\xi_\ell^a = \xi_\ell^d + \varepsilon_\ell^a$$

Arrival delay

Arrival root delay

Departure delay

Delay propagation along aircraft routes



Modeling intrinsic delays

- **Root delays** modeled as (log-)normal distributions:

$$\varepsilon_{\ell}^a \sim \mathcal{N}(\mu_{\ell}^a, \sigma_{\ell}^a),$$

$$\varepsilon_{\ell}^d \sim \log \mathcal{N}(\mu_{\ell}^d, \sigma_{\ell}^d)$$

Any other **differentiable** distribution can be used instead.

Modeling intrinsic delays

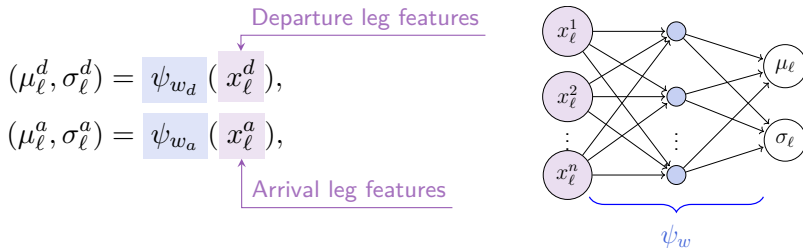
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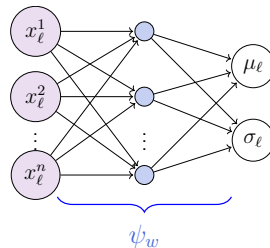
Any other **differentiable** distribution can be used instead.

- Distribution parameters are parameterized by **neural networks**:

$$\begin{aligned} (\mu_{\ell}^d, \sigma_{\ell}^d) &= \psi_{w_d} (x_{\ell}^d), \\ (\mu_{\ell}^a, \sigma_{\ell}^a) &= \psi_{w_a} (x_{\ell}^a), \end{aligned}$$

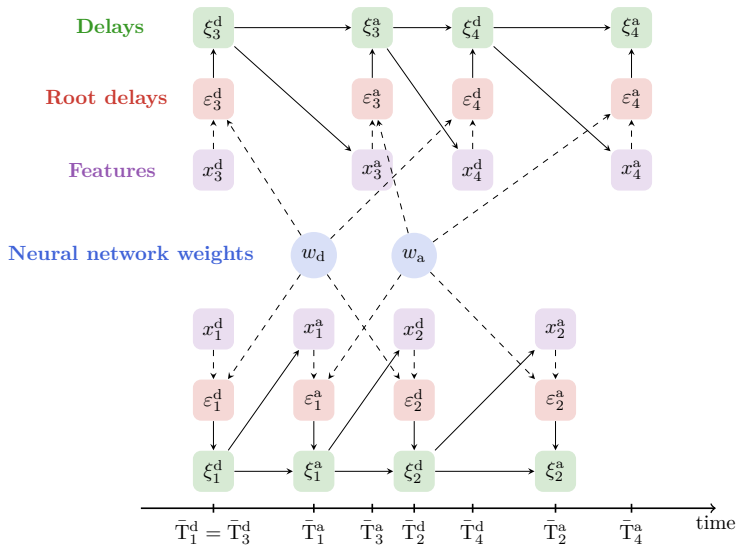
Departure leg features

Arrival leg features



- **Goal:** how to learn parameters w_d and w_a ?

Delay prediction model



Log-likelihood maximization

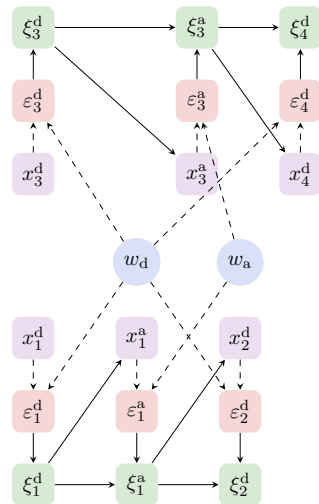
When the Bayesian Network is an **acyclic graph**:

$$p(v_1, \dots, v_n) = \prod_{i=1}^n p(v_i \mid \text{parents}(v_i))$$

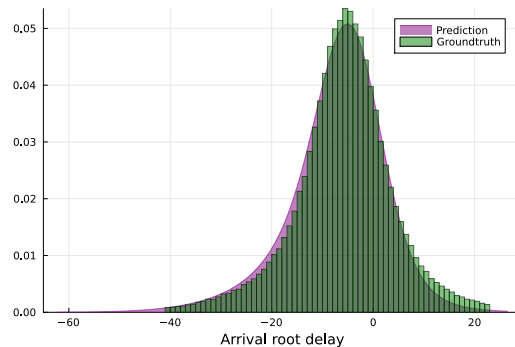
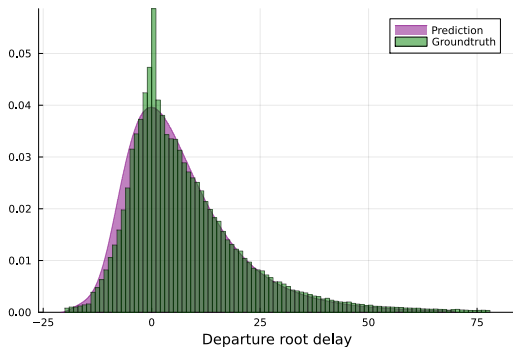
Log-likelihood of neural network parameters:

$$l(w; x, \varepsilon) = \sum_{\ell \in \mathcal{L}} \log p(\varepsilon_{\ell}^d | x_{\ell}^d, w) + \log p(\varepsilon_{\ell}^a | x_{\ell}^a, w)$$

$\Rightarrow$ **gradient descent** optimization can be used



Results



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Compact MIP

c^0 is **non-linear** in the leg-level variables y_a^i :

- delay propagation along a route is non-linear
- delay cost is piecewise linear

Total cost, non-linear (delay propagation)

$$\min_y \quad \frac{1}{S} \sum_{s=1}^S c^0(y; x, \xi_s)$$

$$\text{s.t.} \quad \sum_{a \in \delta^-(v) \cap \mathcal{A}^i} y_a^i = \sum_{a \in \delta^+(v) \cap \mathcal{A}^i} y_a^i, \forall v \in \mathcal{V}, \forall i \in \mathcal{I},$$

$$\sum_{a \in \delta^+(o) \cap \mathcal{A}^i} y_a^i = \sum_{a \in \delta^-(t) \cap \mathcal{A}^i} y_a^i = 1, \forall i \in \mathcal{I},$$

$$\sum_{i \in \mathcal{I}} \sum_{a \in \delta^-(\ell) \cap \mathcal{A}^i} y_a^i = 1, \quad \forall \ell \in \mathcal{L},$$

$$y_a^i \in \{0, 1\}, \forall i \in \mathcal{I}, \forall a \in \mathcal{A}^i.$$

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 & \quad \sum_{a \in \delta^+(o) \cap \mathcal{A}^i} y_a^i = \sum_{a \in \delta^-(t) \cap \mathcal{A}^i} y_a^i = 1, \forall i \in \mathcal{I}, \\
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 \end{aligned}$$

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Flow constraints

Set-partitioning constraints

Compact MIP

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⇒ an exact compact MILP exists (per-scenario delay variables, linearized propagation), but needs $\sim$ **arcs** $\times$ **scenarios** extra variables and has a **weak LP relaxation**: does not scale past a few hundred legs, even with 1 scenario.

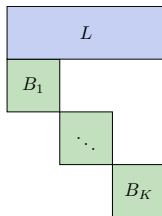
$$\begin{aligned}
 & \min_y \quad \frac{1}{S} \sum_{s=1}^S \underbrace{c^0(y; x, \xi_s)}_{\text{Total cost, non-linear (delay propagation)}} \\
 & \text{s.t.} \quad \underbrace{\sum_{a \in \delta^-(v) \cap \mathcal{A}^i} y_a^i = \sum_{a \in \delta^+(v) \cap \mathcal{A}^i} y_a^i, \forall v \in \mathcal{V}, \forall i \in \mathcal{I}}_{\text{Flow constraints}}, \\
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 & \quad \sum_{i \in \mathcal{I}} \sum_{a \in \delta^-(\ell) \cap \mathcal{A}^i} y_a^i = 1, \quad \forall \ell \in \mathcal{L}, \\
 & \quad y_a^i \in \{0, 1\}, \forall i \in \mathcal{I}, \forall a \in \mathcal{A}^i.
 \end{aligned}$$

Set-partitioning constraints

Route reformulation

Dantzig-Wolfe decomposition: one variable per feasible **route** $r \in \mathcal{R}^i$ instead of one per leg.

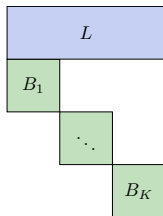
- route cost c_r^i hides the non-linearity:
computed by **simulation** over scenarios,
once the route is fixed
- deals with delay propagation exactly



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Master problem:

$$\min_y \sum_{i \in \mathcal{I}} \sum_{r \in \mathcal{R}^i} c_r^i y_r^i$$

Each leg covered exactly once

$$\text{s.t.} \quad \sum_{i \in \mathcal{I}} \sum_{r \ni \ell, r \in \mathcal{R}^i} y_r^i = 1, \quad \forall \ell \in \mathcal{L},$$

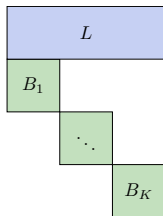
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Route reformulation

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$|\mathcal{R}^i|$ is **exponential** in the number of legs $\implies$ generate routes on the fly instead of enumerating them. Column generation solves the **LP relaxation** of the master (replace $y_r^i \in \{0, 1\}$ by $y_r^i \geq 0$).

Master problem:

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Each leg covered exactly once

$$\begin{aligned} \text{s.t.} \quad & \sum_{i \in \mathcal{I}} \sum_{r \ni \ell, r \in \mathcal{R}^i} y_r^i = 1, & \forall \ell \in \mathcal{L}, \\ & \sum_{r \in \mathcal{R}^i} y_r^i = 1, & \forall i \in \mathcal{I}, \\ & y_r^i \in \{0, 1\}, & \forall i \in \mathcal{I}, \forall r \in \mathcal{R}^i. \end{aligned}$$

Easier to see on the dual: row generation

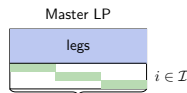
Dual of the master LP relaxation: one variable per leg and per aircraft, one constraint per route.

$|\mathcal{L}| + |\mathcal{I}|$ variables: few

$$\max_{\lambda, \mu} \quad \sum_{\ell \in \mathcal{L}} \lambda_{\ell} + \sum_{i \in \mathcal{I}} \mu_i$$

exponentially many

$$\text{s.t.} \quad \sum_{\ell \in r} \lambda_{\ell} + \mu_i \leq c_r^i, \quad \forall i \in \mathcal{I}, \forall r \in \mathcal{R}^i.$$



routes: exp. many columns

Easier to see on the dual: row generation

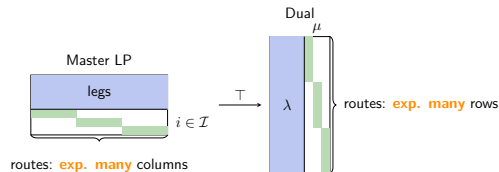
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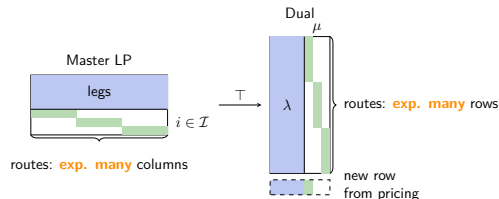
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$$\text{s.t.} \quad \sum_{\ell \in r} \lambda_{\ell} + \mu_i \leq c_r^i, \quad \forall i \in \mathcal{I}, \forall r \in \mathcal{R}^i.$$

Row generation: solve the dual over a few routes, read (λ, μ) , add the **most violated** constraint for each aircraft:

$$\min_{r \in \mathcal{R}^i} c_r^i - \sum_{\ell \in r} \lambda_{\ell} - \mu_i < 0, \quad i \in \mathcal{I}$$

$\iff$ a route with negative reduced cost.



Easier to see on the dual: row generation

Dual of the master LP relaxation: one variable per leg and per aircraft, one constraint per route.

$|\mathcal{L}| + |\mathcal{I}|$ variables: few

$$\max_{\lambda, \mu} \sum_{\ell \in \mathcal{L}} \lambda_{\ell} + \sum_{i \in \mathcal{I}} \mu_i$$

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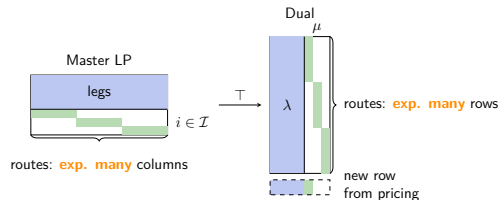
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$\iff$ a route with negative reduced cost.

Stop when no constraint is violated: (λ, μ) is feasible for the full dual, so the LP relaxation is solved.



In practice: adding a **constraint** is one line in a modeling language, adding a column is not.

Column generation: the algorithm

initialize $\tilde{\mathcal{R}}^i$ with a few feasible routes for each $i \in \mathcal{I}$;
repeat
 solve the restricted master **LP** (or its dual) over the
 current sets $(\tilde{\mathcal{R}}^i)_{i \in \mathcal{I}}$;
 get duals λ_ℓ ($\ell \in \mathcal{L}$) and μ_i ($i \in \mathcal{I}$);

until *no route with negative reduced cost is found*;

Column generation: the algorithm

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  solve the restricted master LP (or its dual) over the  
    current sets  $(\tilde{\mathcal{R}}^i)_{i \in \mathcal{I}}$ ;  
  get duals  $\lambda_\ell$  ( $\ell \in \mathcal{L}$ ) and  $\mu_i$  ( $i \in \mathcal{I}$ );  
  for each aircraft  $i \in \mathcal{I}$  do  
    pricing (most violated dual constraint):  
       $r^* \in \operatorname{argmin}_{r \in \mathcal{R}^i} c_r^i - \sum_{\ell \in r} \lambda_\ell - \mu_i$ ;  
      if the minimum is  $< 0$  then  
        | add  $r^*$  to  $\tilde{\mathcal{R}}^i$ ;  
      end  
  end  
until no route with negative reduced cost is found;
```

- **Pricing**: finds the best route for the current prices *without* enumerating $\mathcal{R}^i$: this is the problem-specific difficult part

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        if the minimum is  $< 0$  then  
            add  $r^*$  to  $\tilde{\mathcal{R}}^i$ ;  
        end  
    end  
until no route with negative reduced cost is found;  
return LP optimum (lower bound) and fractional  $y$ 
```

- **Pricing**: finds the best route for the current prices *without* enumerating $\mathcal{R}^i$: this is the problem-specific difficult part
- **Stopping rule**: no negative reduced cost means no violated dual constraint, so the restricted LP optimum is optimal for the full master

Pricing: a stochastic shortest path

The diagram shows the pricing problem equation:
$$\min_{r \in \mathcal{R}^i} c_r^i - \sum_{\ell \in r} \lambda_\ell - \mu_i$$
 Annotations include:

- A red line from "expected delay cost, non-linear" to the c_r^i term.
- A green line from "constant" to the μ_i term.
- A blue line from "leg prices: additive, one per arc" to the summation term $\sum_{\ell \in r} \lambda_\ell$.

One problem per aircraft $i \in \mathcal{I}$. The route r^* is added as a new column (a new row of the dual) when the minimum is negative.

Pricing: a stochastic shortest path

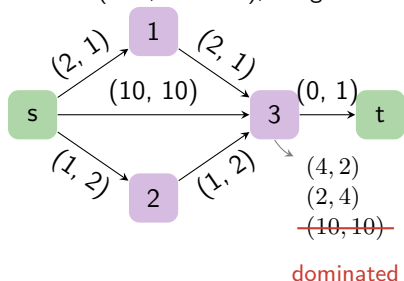
The diagram shows the pricing problem equation:
$$\min_{r \in \mathcal{R}^i} c_r^i - \sum_{\ell \in r} \lambda_\ell - \mu_i$$
 Annotations: A red line points from "expected delay cost, non-linear" to the c_r^i term (highlighted in a pink box). A green line points from "constant" to the μ_i term (highlighted in a green box). A blue line points from "leg prices: additive, one per arc" to the summation term $\sum_{\ell \in r} \lambda_\ell$ (highlighted in a blue box).

One problem per aircraft $i \in \mathcal{I}$. The route r^* is added as a new column (a new row of the dual) when the minimum is negative.

With the leg prices alone this is a **plain shortest path** in the flight graph. The delay **propagates** along the route, so c_r^i is a **non-additive resource**, one per scenario: a constrained shortest path.

Pricing: label-setting with dominance and bounding

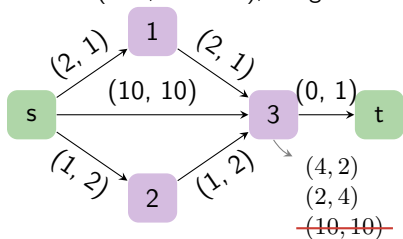
arc labels (cost, resource), budget $W = 4$



- **Enumeration** (label-setting): extend partial paths from s , one label (cost, resource) per partial path
- **Dominance**: at node 3, $(10, 10)$ is worse than $(4, 2)$ on both, drop it. Keep $(4, 2)$ and $(2, 4)$: neither dominates the other, and the cheaper one cannot reach t within the budget

Pricing: label-setting with dominance and bounding

arc labels (cost, resource), budget $W = 4$



dominated

- **Enumeration** (label-setting): extend partial paths from s , one label (cost, resource) per partial path
- **Dominance**: at node 3, (10, 10) is worse than (4, 2) on both, drop it. Keep (4, 2) and (2, 4): neither dominates the other, and the cheaper one cannot reach t within the budget
- **Bounding**: a backward pass gives a lower bound on the cost from each node to t , prune labels that cannot beat the best route found
- **Our pricing**: cost is the reduced cost, the resource is the propagated delay in each scenario, handled as one **convex piecewise linear function** per scenario

From fractional to integer: restricted master heuristic

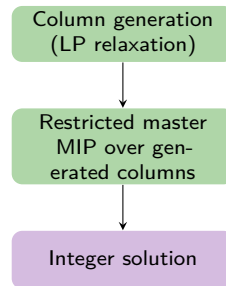
Column generation returns a fractional LP solution y (with the **LP relaxation lower bound**): we need an integer one.

Column generation
(LP relaxation)

From fractional to integer: restricted master heuristic

Column generation returns a fractional LP solution y (with the **LP relaxation lower bound**): we need an integer one.

Restricted master heuristic: keep the columns generated so far, restore $y_r^i \in \{0, 1\}$, and solve as a **MIP** with an off-the-shelf solver.

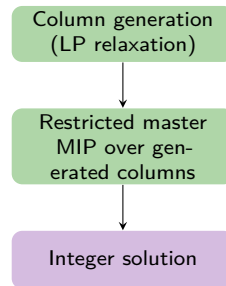


From fractional to integer: restricted master heuristic

Column generation returns a fractional LP solution y (with the **LP relaxation lower bound**): we need an integer one.

Restricted master heuristic: keep the columns generated so far, restore $y_r^i \in \{0, 1\}$, and solve as a **MIP** with an off-the-shelf solver.

Pros: simple, cheap in principle (columns are already generated), any feasible solution gives an **upper bound**.



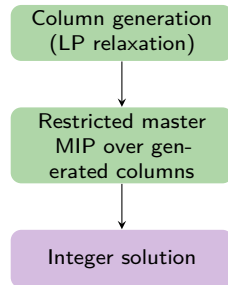
From fractional to integer: restricted master heuristic

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Restricted master heuristic: keep the columns generated so far, restore $y_r^i \in \{0, 1\}$, and solve as a **MIP** with an off-the-shelf solver.

Pros: simple, cheap in principle (columns are already generated), any feasible solution gives an **upper bound**.

Cons: the columns were selected to solve the LP, not to combine into an integer solution $\implies$ a good integer solution may simply not be in the pool.



Restricted master heuristic: results

Air France instances, 60 to 600 flight legs, 129 instances. Gaps are relative to the **column generation lower bound**. Times are capped by a **one-hour time limit**.

Scenarios	Column generation	Restricted master heuristic	
	Average time	Average time	Average gap
1	578s	379s	74%
5	667s	654s	137%
10	915s	780s	147%
50	1596s	638s	143%
100	2738s	568s	117%

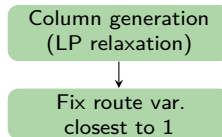
Cheap, but the generated columns do not combine into good integer solutions: **gaps above 70%**.

From fractional to integer: diving heuristic

Diving: **depth-first** descent in the branch-and-price tree, with **limited backtracking**.

At each node:

1. solve the LP by column generation
2. fix the route variable y_r^i with the largest fractional value to 1, removing aircraft i and its legs from the graph

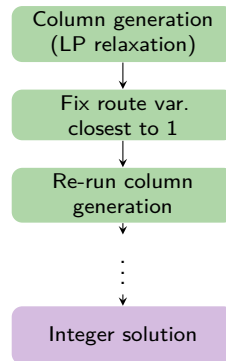


From fractional to integer: diving heuristic

Diving: **depth-first** descent in the branch-and-price tree, with **limited backtracking**.

At each node:

1. solve the LP by column generation
2. fix the route variable y_r^i with the largest fractional value to 1, removing aircraft i and its legs from the graph
3. re-run column generation on the reduced problem, reusing the still-compatible columns
4. if a node is infeasible, apply **limited backtracking**: try the next-largest fractional route instead, up to a taboo list of size k ($k = 15$)
5. repeat until the solution is integer (at most $|\mathcal{I}|$ fixings)



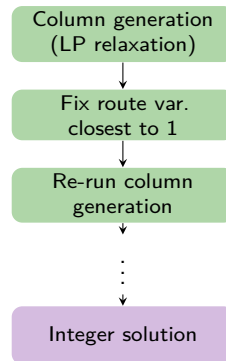
From fractional to integer: diving heuristic

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5. repeat until the solution is integer (at most $|\mathcal{I}|$ fixings)

Shrinking the graph makes column generation **cheaper** at deeper nodes: on average diving costs about one to two column generation runs.



Diving heuristic: results

Air France instances, 60 to 600 flight legs, 129 instances. Gaps are relative to the **column generation lower bound**. Times are capped by a **one-hour time limit**.

Scenarios	Column generation	Restricted master heuristic		Diving heuristic	
	Avg. time	Avg. time	Avg. gap	Avg. time	Avg. gap
1	578s	379s	74%	870s	0.1%
5	667s	654s	137%	1381s	0.2%
10	915s	780s	147%	818s	0.4%
50	1596s	638s	143%	1338s	0.2%
100	2738s	568s	117%	2061s	0.3%

About the cost of one to two column generation runs, and **gaps of a few tenths of a percent**: near-optimal at Air France scale.

Branch-and-price: the exact alternative

- Branch-and-bound where **column generation is solved at every node**
- Branching rule must stay compatible with the pricing problem: branch on **arcs of the flight graph**, not on route variables y_r^i
 - ▶ fixing $y_r^i = 1$ (as diving does) is compatible with pricing: it removes aircraft i and its legs from the graph
 - ▶ a branch $y_r^i = 0$ would require excluding one specific path from the pricing, which label-setting cannot express
- Gives an **optimality proof** (exact method), but cost grows with the size of the search tree

	Exact?	Computational cost	Solution quality
Restricted master heuristic	✗	low (but hits the time limit on large instances)	poor (large gap)
Diving	✗	moderate	near-optimal
Branch-and-price	✓	high (tree size)	optimal

⇒ **Diving is the practical choice for tail assignment.**

What we have now

- **Near-optimal solutions** (a few tenths of a percent from the column generation lower bound), but **hours per instance**, growing with the number of legs and scenarios
- A **lower bound** from the column generation (LP) relaxation
- The ability to build a **dataset** of (instance, solution) pairs
- $\implies$ used in part 2 to **learn** a faster decision-making policy

Practice session 1

See 02_or_notebook.jl